The Metric Is Not the Construct

Construct validity at every layer of a lending audit

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1 THE QUESTION

Models now decide consumer installment credit and residential mortgage origination — a setting where the constructs an audit must measure come with written definitions and published flags. **At each layer of a fairness audit — the metric library, the mapping from attribute to construct, and the audit's own evaluation instruments — does the quantity actually computed match the construct it is named for, and would anyone be able to tell from the output?**

2 WHY IT MATTERS

Five decisions determine whether an audit's numbers mean what its readers take them to mean: which outcome counts as favorable, who counts as protected, against whom the comparison runs, how much evidence supports each subgroup estimate, and which instruments check the audit's own components. Get any one wrong and the audit still runs, still emits a number, and still reads clean — the direction nobody re-examines. A false alarm costs a reviewer an afternoon; a false clearance closes the question and the applicant never learns why.

3 WHAT WE DID

A complete audit of both sectors on real data: 1,000 consumer credit and 10,978 Georgia mortgage applications, every applicant scored out-of-fold by a model that never saw them. Deterministic code owns every number (AIF360 computes the metrics, a fixed rule assigns severity); a language model owns only interpretation and is scored against the deterministic label (98 calls, $0.89, offline replay without an API key). We then audited the audit: a subsampling study — 2,000 draws at each of five audit sizes, five reporting policies, three reference-group conventions against the full-population table — and a reanalysis of every automated check our own harness applies to model output.

4 HOW IT IS RELEVANT

The library layer. Verified directly against AIF360 0.6 and Fairlearn 0.13: one AIF360 constructor defaults `favorable_label=1.0`, so on a target named `loan_default` it reports the disadvantaged group as *favoured* (1.2072 vs. the oriented 0.9862); and neither library ships an interval, an n, or a record of the reference group with a disaggregated ratio.

The definition layer. Auditing age at 40+ — a threshold imported from employment practice — returns 0.9791: clean. At 62+, the boundary the mortgage data ship a dedicated flag for, the same model over the same 10,978 rows returns **0.8897**, the largest age disparity present. And at the audit size we originally used (m=2,196), the subsampling study shows a no-floor audit falsely clears a violating race×sex cell in **83.8%** of draws, while reporting floors never do — because they misidentify the worst-treated group in 100% of draws by never reporting it.

The instrument layer. A refusal detector operationalising "refusal" as *under 40 characters* flagged six correct terse answers and contributed to permanently blocking a provider; a fabrication detector flagged arithmetic (0.8888 restated as 88.88%); and a reported cross-model prompt-sensitivity spread (14.7 vs. 0.0 points) concentrates in the one prompt cycle whose correct answers are terse — excluding it, 6.7 — so the published magnitude was partly our own scorer.

5 HOW IT APPLIES

1. State the favorable outcome explicitly; never accept the default. **2.** Report the interval, the n, and the reference group beside every ratio, and publish what any floor suppressed. **3.** Put the definition's source beside the threshold in code — a comment asserting the wrong provenance for 40+ survived review because it looked authoritative. **4.** Treat every mechanical gate as a hypothesis to falsify, starting where the correct answer is unusual. **5.** Audit the book, not a test split, and disclose population construction. **6.** Name the reporting policy — floor, interval, or shrinkage is a choice among failure modes: silence, abstention, or confident clearance.

6 THE PROOF

Table 1: Every claim regenerates offline from the released repository.

Claim	Where to check it
AIF360 default inverts a risk target	pinned repro: default 1.2072 vs. oriented 0.9862; `scripts/aif360_default_repro.py`
False clearance 83.8% at m=2,196; floors wrong 100%	`scripts/subsample_study.py`
6/12 cells band-indeterminate; 0.6734 $[0.45, 0.90]$ at n=31	`scripts/interval_estimates.py`
Banded 0.9791 vs. 62+ 0.8897 (same 10,978 applicants)	`hmda/fairness/fairness_summary.json`
45.1% attrition; Race Not Available = 20.7% of raw file	`scripts/hmda_attrition.py`
Spread 14.7 → 6.7 excl. the terse cycle	`scripts/recheck_prompt_sensitivity.py`

86 tests, CI green; the deterministic layer regenerates with no API key.